

Keeping Time: Day, Month, Year and Calendars

The spinning Earth gives the day, the Moon's cycle the month, and Earth's orbit the year — the basis of every calendar.

Natural cycles become units of time

Unit	Natural cycle	Duration
Day	Earth's rotation on its axis	~24 hours (mean solar day)
Month	Moon's cycle of phases (revolution around Earth)	~29.5 days
Year	Earth's revolution around the Sun (one cycle of seasons)	~365¼ days

The shadow clock

The Sun is **highest** in the sky at midday, when shadows are **shortest**. The time from one such highest point to the next is the **mean solar day** (~24 hours). A **sundial** uses a shadow to tell time.

Calendars

- **Lunar calendar** — based on the Moon's phases; ~12 lunar months fit in one cycle of seasons.
- **Solar calendar** — based on the year (Earth's revolution); the one we commonly use.
- Many Indian calendars are **lunisolar** — they combine the Moon's months with the solar year.

Why a leap year? A year is about 365¼ days, so we add an extra day (Feb 29) every 4 years to keep the calendar in step with the seasons.

Where students lose marks

Trap 1 — Day = rotation; year = revolution. Earth spinning on its axis gives the day; Earth going around the Sun gives the year.

Trap 2 — A month is ~29.5 days (Moon's phase cycle), not exactly 30.

Trap 3 — Shortest shadow = midday (Sun highest). Not sunrise or sunset.

Model answers — write them like this

Example A. Which motion of the Earth gives us the day, and which the year?

Step 1: Earth's rotation on its axis → day (~24 h).

Step 2: Earth's revolution around the Sun → year (~365¼ days).

∴ Rotation = day; revolution = year.

Example B. When is a stick's shadow shortest, and what does it mark?

Step 1: The shadow is shortest when the Sun is highest (midday).

∴ The shortest shadow marks local noon — the basis of the mean solar day.

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